Griffith UNIVERSITY

3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture 5: Mining Frequent Patterns II



Dr Can Wang

School of Information and Communication Technology

Mining Frequent Patterns II

- FP-Growth: A Frequent Pattern-Growth Approach
- Multi-level Association Rules Mining
- Multi-dimensional Association Rules Mining

Bottlenecks of Apriori Approach

Griffith UNIVERSITY

- Bottlenecks of the Apriori approach
  - Need to generate candidate (k+1)-itemsets after counting the support to find out frequent k-itemsets. This candidate generation could be very EXPENSIVE.
    - E.g. 10^4 frequent 1-itemsets → over 10^7 candidate 2-itemsets.
  - Repeatedly scanning the transaction database for support counting dramatically slows down the Apriori algorithm.

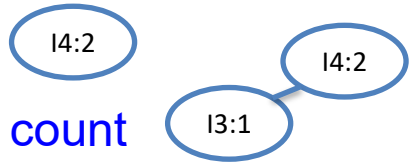
Pattern-Growth Approach: Mining Frequent Patterns Without Candidate Generation

- Bottlenecks of the Apriori approach
  - Breadth-first (i.e., level-wise) search
  - Candidate generation and test
    - Often generates a huge number of candidates
- The FP-Growth Approach (J. Han, J. Pei, and Y. Yin, SIGMOD' 00)
  - Depth-first search
  - Avoid explicit candidate generation

FP-growth: Frequent Pattern-Growth

Griffith UNIVERSITY

- Adopts a divide and conquer strategy
- Compresses the database to represent frequent items into a frequent–pattern tree or FP-tree
  - Retains the itemset association information
  - Each node is represented by an item and its support count
  - Each branch is a k-itemset (or an association) with support count



- Once the FP-tree is constructed, mine the frequent itemsets from the FP-tree
  - Divide the FP-tree into different parts and perform the mining over the parts
- Divides the compressed database into a set of conditional databases, each associated with one frequent item
- Mines each such database separately

Example: FP-growth

- The first scan of data is the same as Apriori
- Derive the set of frequent 1-itemsets
- Let min-sup-count=2
- Generate a set of ordered items

Transactional Database

TID	List of item IDS
T100	I1,I2,I5
T200	I2,I4
T300	I2,I3
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I2,I3

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2

Construct the FP-Tree

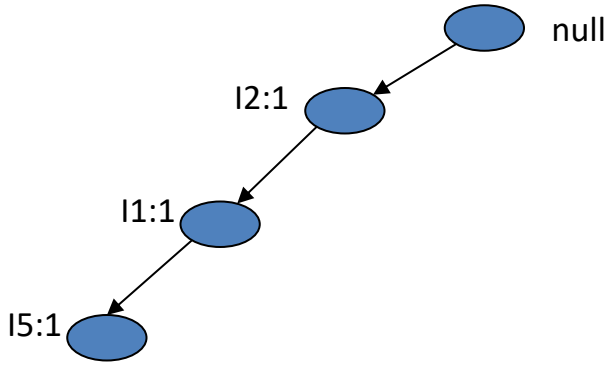
Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

- Create a branch for each transaction
- Items in each transaction are processed in order

1- Order the items T100: {I2,I1,I5}
2- Construct the first branch: <I2:1>, <I1:1>,<I5:1>

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2



Construct the FP-Tree

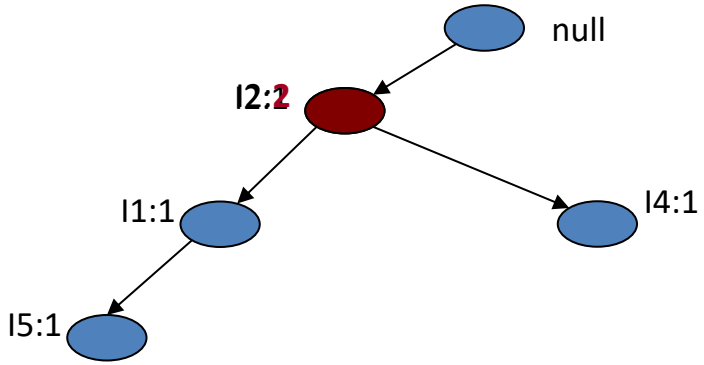
Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

- Create a branch for each transaction
- Items in each transaction are processed in order

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2

1- Order the items T200: {I2,I4}
2- Construct the second branch: <I2:1>, <I4:1>



Construct the FP-Tree

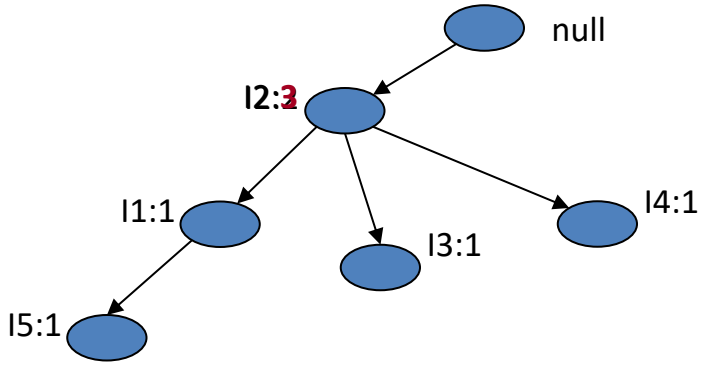
Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

- Create a branch for each transaction
- Items in each transaction are processed in order

1- Order the items T300: {I2,I3}
2- Construct the third branch: <I2:2>, <I3:1>

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2



Construct the FP-Tree

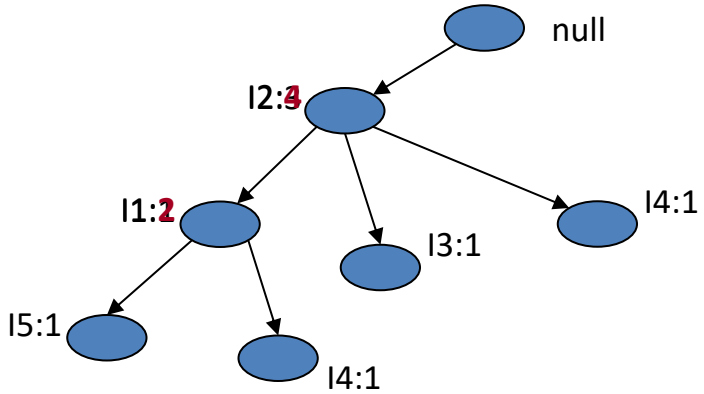
Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

- Create a branch for each transaction
- Items in each transaction are processed in order

1- **Order the items T400: {I2,I1,I4}**
2- **Construct the fourth branch: <I2:3>, <I1:1>,<I4:1>**

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2



Construct the FP-Tree

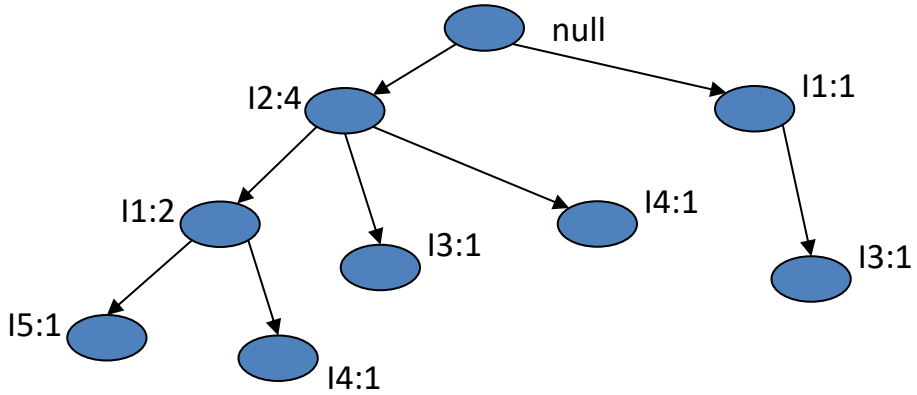
Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

- Create a branch for each transaction
- Items in each transaction are processed in order

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2

1- Order the items T500: {I1,I3}
2- Construct the fifth branch: <I1:1>, <I3:1>

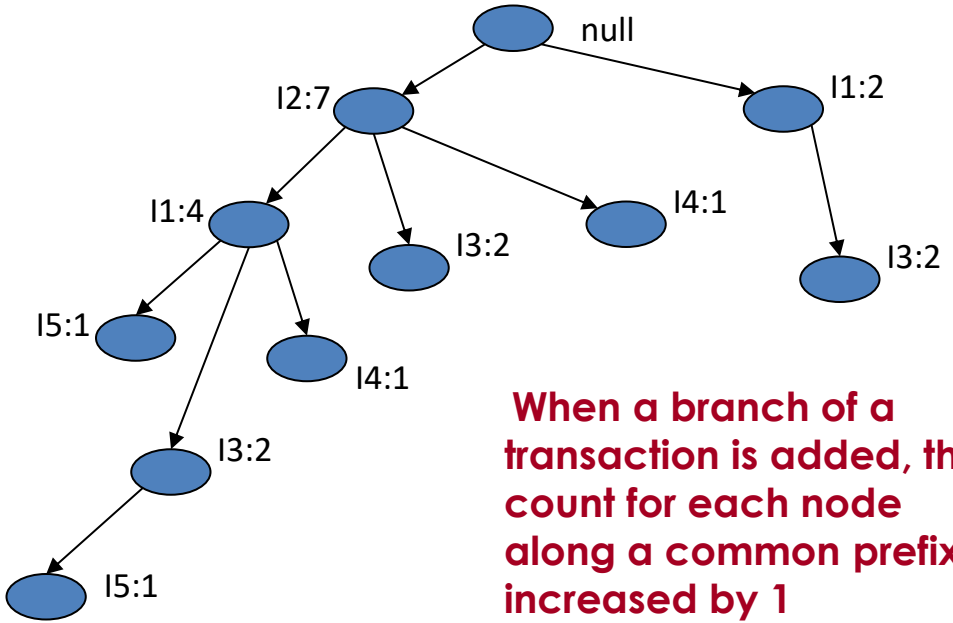


Construct the FP-Tree

Transactional Database

TID	Items	TID	Items	TID	Items
T100	I1,I2,I5	T400	I1,I2,I4	T700	I1,I3
T200	I2,I4	T500	I1,I3	T800	I1,I2,I3,I5
T300	I2,I3	T600	I2,I3	T900	I1,I2,I3

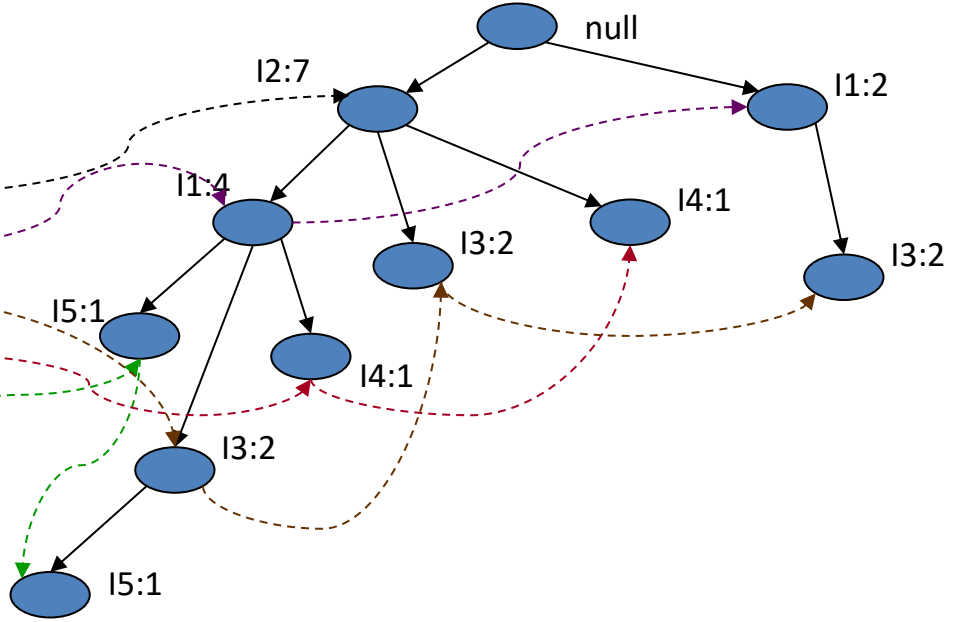
Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2



When a branch of a transaction is added, the count for each node along a common prefix is increased by 1

Construct the FP-Tree

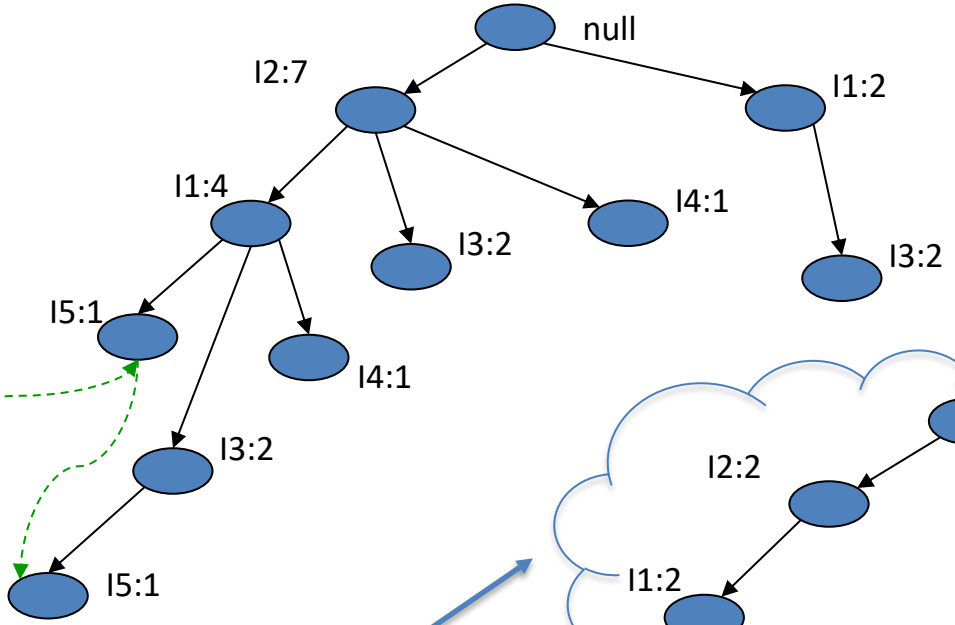
Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2

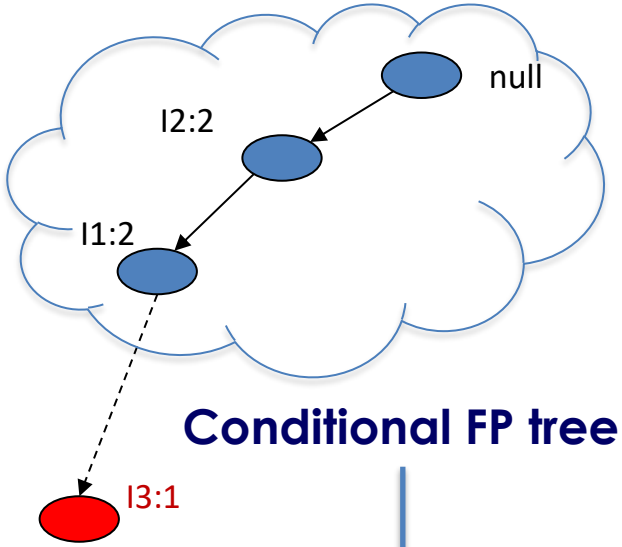


The problem of mining frequent patterns in databases is transformed to that of mining the FP-tree

Mining FP-Tree for Frequent Patterns

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2





Conditional FP tree

-Occurrences of I5: <I2, I1, I5> and <I2, I1, I3, I5>

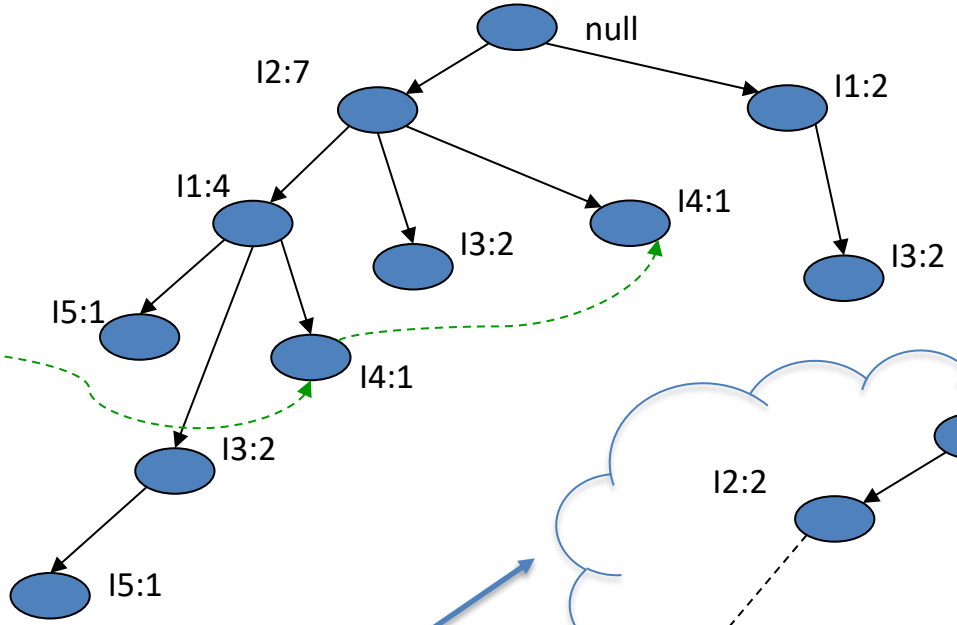
-Two prefix Paths <I2, I1: 1> and <I2, I1, I3: 1>

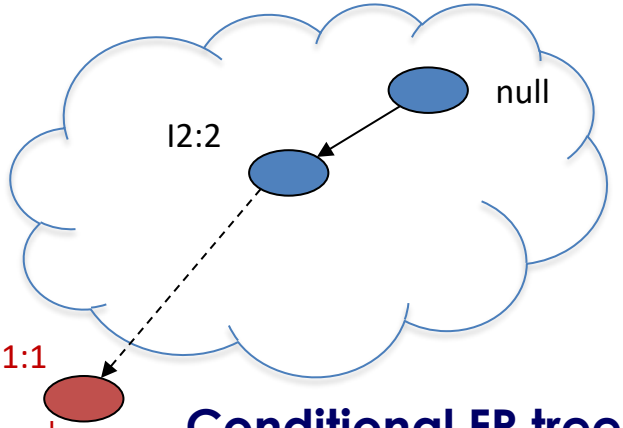
-Conditional FP tree contains only <I2: 2, I1: 2>, I3 is not considered because its support count of 1 is less than the minimum support count.

-Frequent patterns {I2, I5: 2}, {I1, I5: 2}, {I2, I1, I5: 2}

Mining FP-Tree for Frequent Patterns

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2





Conditional FP tree

-Occurrences of I4: <I2, I1, I4> and <I2, I4>

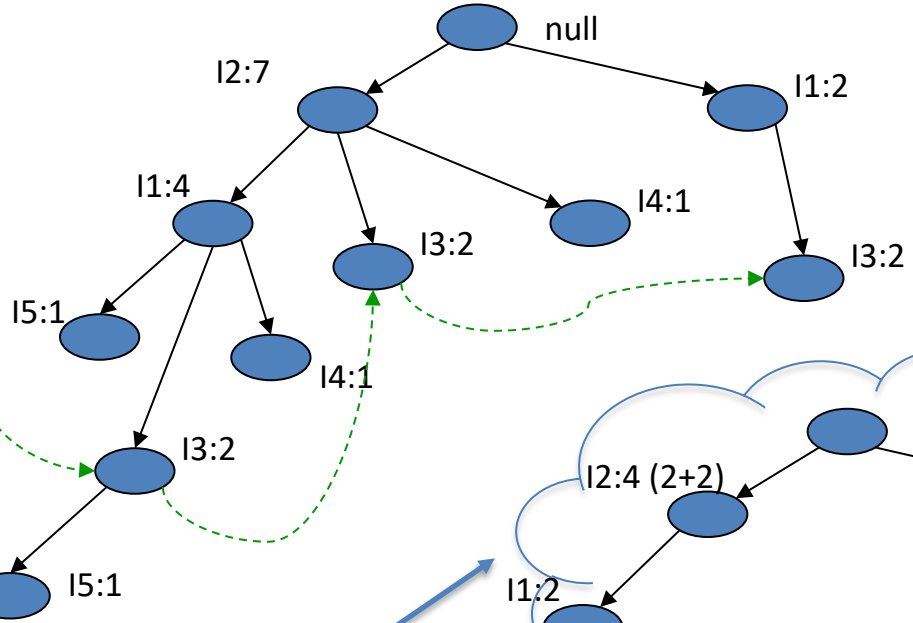
-Two prefix Paths <I2, I1: 1> and <I2: 1>

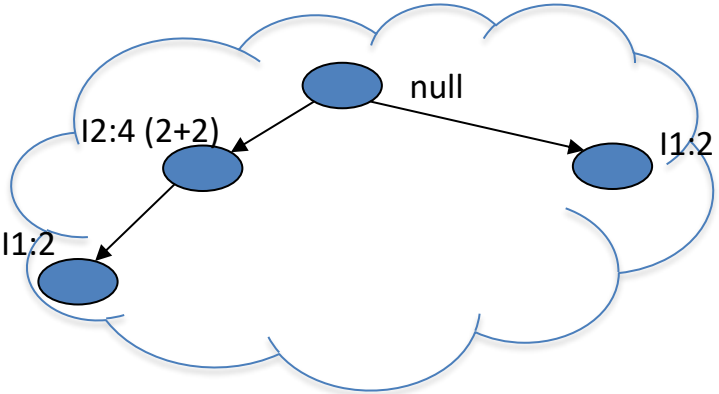
-Conditional FP tree contains only <I2: 2>, I1 is not considered because its support count of 1 is less than the minimum support count.

-Frequent patterns {I2, I4: 2}

Mining FP-Tree for Frequent Patterns

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2





Conditional FP tree

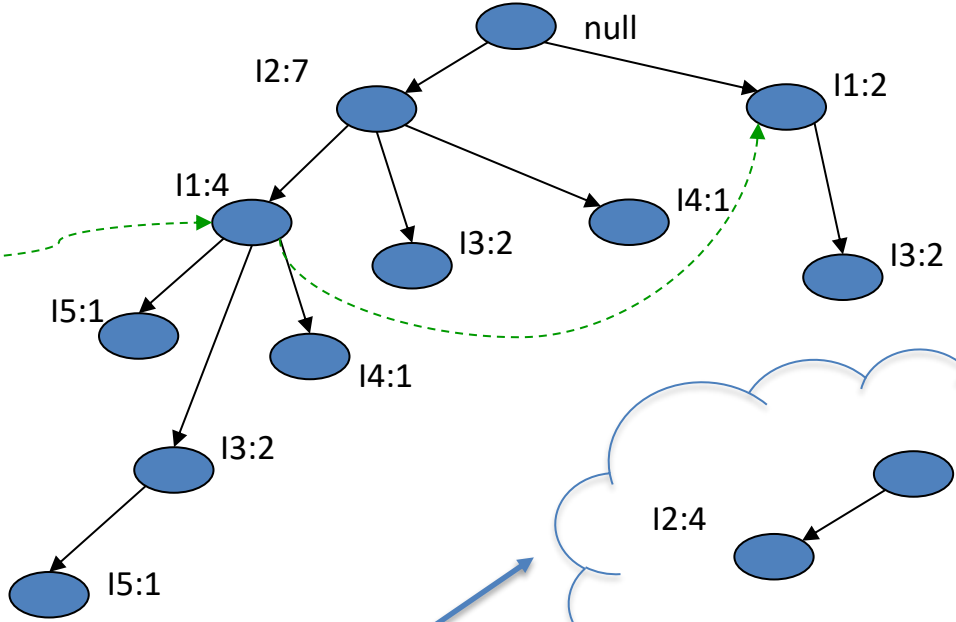
-Occurrences of I3: <I2, I1, I3>, <I2, I3>, and <I1, I3>

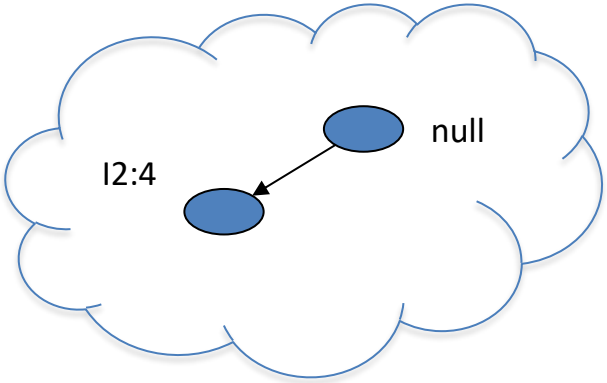
-Three prefix Paths <I2, I1: 2>, <I2: 2>, and <I1: 2>

-Frequent patterns {I2, I3: 4}, {I1, I3: 4}, {I2, I1, I3: 2}

Mining FP-Tree for Frequent Patterns

Item ID	Support count
I2	7
I1	6
I3	6
I4	2
I5	2





Conditional FP tree

-Occurrences of I1: <I2, I1>

-One prefix Path <I2: 4>

-Conditional FP tree contains only <I2: 4>, I1 is not considered because there is only one item I2 as a **prefix**.

-Frequent patterns {I2, I1: 4}

Summary of FP-Growth

TID	Conditional FP-tree	Frequent Patterns Generated
I5	<I2:2,I1:2>	{I2,I5:2}, {I1,I5:2},{I2,I1,I5:2}
I4	<I2:2>	{I2,I4:2}
I3	<I2:4,I1:2>,<I1:2>	{I2,I3:4},{I1,I3:4},{I2,I1,I3:2}
I1	<I2:4>	{I2,I1:4}

- FP-growth transforms the problem of finding long frequent patterns to searching for shorter ones recursively and concatenating the suffix.
- It uses the least frequent suffix offering a good selectivity.
- It reduces the search cost.
- If the tree does not fit into main memory, partition the database into a set of projected databases to mine FP-tree there.
- Efficient and scalable for mining both long and short frequent patterns.

Mining Frequent Patterns II

- FP-Growth: A Frequent Pattern-Growth Approach
- Multi-level Association Rules Mining
- Multi-dimensional Association Rules Mining

Mining Multi-level Associations

Griffith UNIVERSITY

- Strong associations discovered at high abstraction levels, though with high support, could be common sense knowledge.
- Finding novel patterns at more detailed levels (like Drill-down) could be interesting.
- Mining patterns at higher levels (like Drill-up) can avoid too many scattered patterns at low or primitive abstraction levels.
- Therefore, it is important to develop effective methods for mining patterns at multiple abstraction levels (Drill-up and Drill-down).

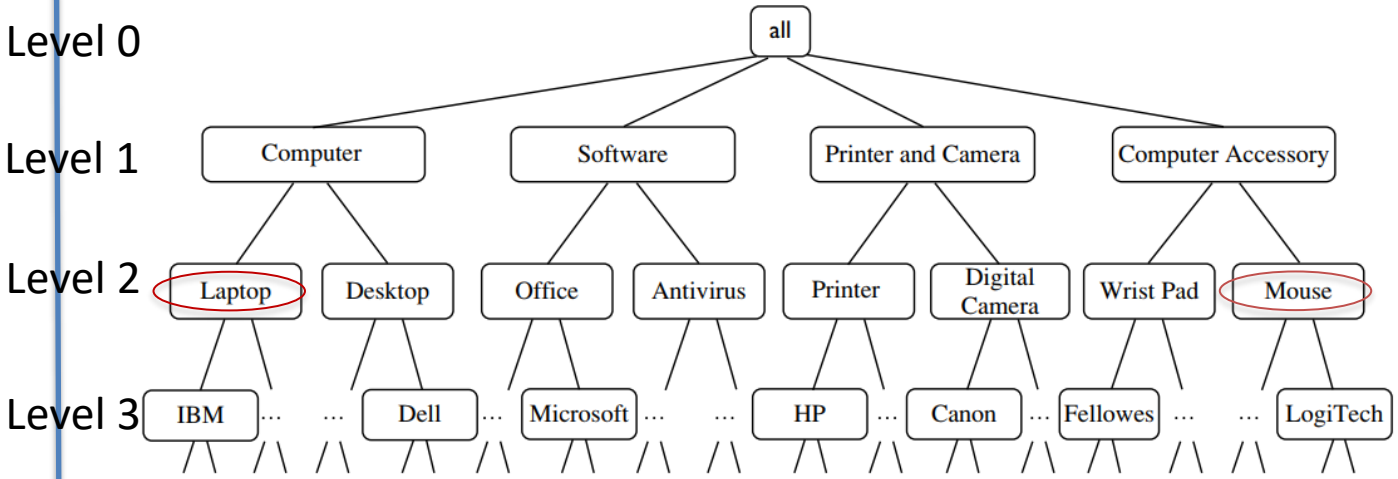
Mining Multi-level Associations

task-relevant set of transactional data

Task-Relevant Data, D

TID	Items Purchased
T100	Apple 17″ MacBook Pro Notebook, HP Photosmart Pro b9180
T200	Microsoft Office Professional 2010, Microsoft Wireless Optical Mouse 5000
T300	Logitech VX Nano Cordless Laser Mouse, Fellowes GEL Wrist Rest
T400	Dell Studio XPS 16 Notebook, Canon PowerShot SD1400
T500	Lenovo ThinkPad X200 Tablet PC, Symantec Norton Antivirus 2010

concept hierarchy for the items



Lowest level

Concept hierarchy for *AllElectronics* computer items.

Mining Multi-level Associations

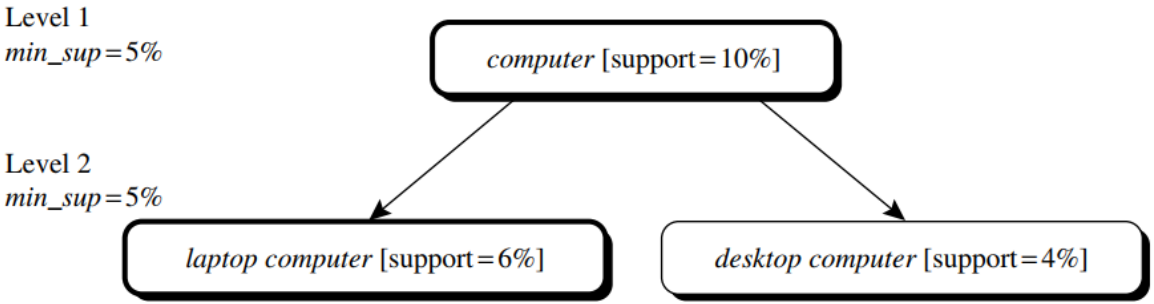
Griffith UNIVERSITY

- Association rules generated from mining data at multiple abstraction levels are called multiple-level or multi-level association rules. Multi-level association rules can be mined efficiently using concept hierarchies under a support-confidence framework.

- In general, **a top-down strategy** is employed, where counts are accumulated for the calculation of frequent itemsets at each concept level, starting at concept level 1 and working downward in the hierarchy toward the more specific concept levels, until no more frequent itemsets can be found.

Mining Multi-level Associations

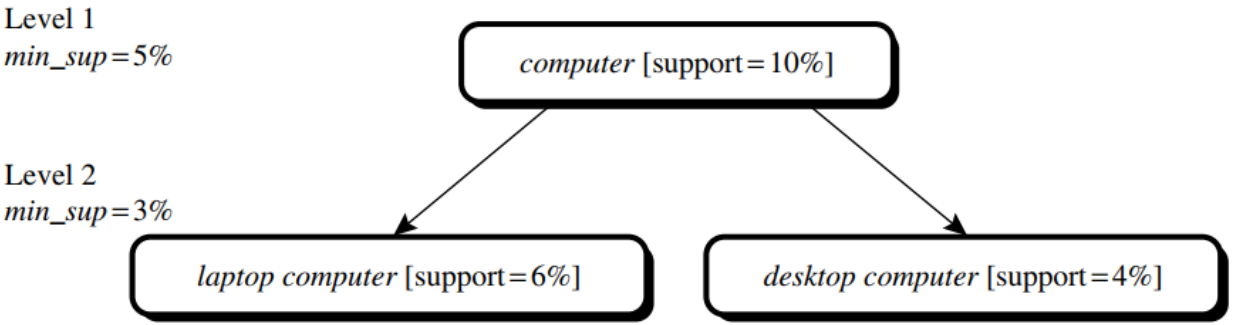
- **Using uniform minimum support for all levels (referred to as uniform support)**
  - The same minimum support threshold is used when mining at each abstraction level.
  - For example, a minimum support threshold of 5% is ONLY used throughout (e.g., for mining from “computer” downward to “laptop computer”).
  - Both “computer” and “laptop computer” are found to be frequent, whereas “desktop computer” is not.
  - **Setting minimum support threshold is very crucial**, as too high or too low will have different results at low/high abstraction levels.



Multilevel mining with uniform support.

Mining Multi-level Associations

- **Each abstraction level has its own minimum support threshold**
  - The deeper the abstraction level, the smaller the corresponding threshold. For example, the minimum support thresholds for levels 1 and 2 are 5% and 3%, respectively. In this way, “computer,” “laptop computer,” and “desktop computer” are all considered frequent.



Multilevel mining with reduced support.

Mining Multi-level Associations

Griffith UNIVERSITY

- **Flexible min-support thresholds:** Some items are more valuable but less frequent
  - Use non-uniform, group-based min-support
  - E.g., {diamond, watch, camera}: 0.05%; {bread, milk}: 5%; …
- **Redundancy Filtering:** Some rules may be redundant due to "ancestor" relationships between items
  - milk → wheat bread [support = 8%, confidence = 70%] (one year)
  - milk → wheat bread [support = 2%, confidence = 72%] (one quarter)
  - The first rule is an ancestor of the second rule (2% = 8% * ¼)
  - A rule is redundant if its support is close to the "expected" value, based on the rule's ancestor

Mining Frequent Patterns II

- FP-Growth: A Frequent Pattern-Growth Approach
- Multi-level Association Rules Mining
- Multi-dimensional Association Rules Mining

Mining Multi-dimensional Associations

Griffith UNIVERSITY

- So far, we have studied association rules that imply a single predicate, that is, the predicate buys. For instance, in mining our AllElectronics database, we may discover the Boolean association rule

$$buys(X, \text{“}digital\ camera\text{”}) \Rightarrow buys(X, \text{“}HP\ printer\text{”}).$$

- we can refer to this rule as a **single dimensional** or **intradimensional association rule,** because it contains a single distinct predicate (e.g., ***buys***) with multiple occurrences (i.e., the predicate occurs more than once within the rule).

Mining Multi-dimensional Associations

Griffith UNIVERSITY

- Instead of considering transactional data only, sales and related information are often linked with relational data or integrated into a data warehouse. Such data stores are multidimensional in nature.
- Considering each database attribute or warehouse dimension as a predicate, we can mine association rules containing multiple predicates:

$$age(X, \text{“}20\ldots29\text{”}) \wedge occupation(X, \text{“}student\text{”}) \Rightarrow buys(X, \text{“}laptop\text{”}).$$

- Association rules that involve two or more dimensions or predicates can be referred to as **multidimensional association rules**.

Mining Multi-dimensional Associations

Griffith UNIVERSITY

- We can also mine multidimensional association rules with **repeated** predicates, which contain multiple occurrences of some predicates. These rules are called **hybrid-dimensional association rules**. An example of such a rule is the following, where the predicate buys is repeated:

$$age(X, \text{“}20\ldots29\text{”}) \wedge buys(X, \text{“}laptop\text{”}) \Rightarrow buys(X, \text{“}HP\ printer\text{”}).$$

Mining Multi-Dimensional Associations

Griffith UNIVERSITY

- Single-dimensional rules:

  buys(X, “milk”) $\Rightarrow$ buys(X, “bread”)

- Multi-dimensional rules: $\geq$ 2 dimensions or predicates
  - Inter-dimension assoc. rules (*no repeated predicates*)
    - age(X,”19-25”) $\wedge$ occupation(X,“student”) $\Rightarrow$ buys(X, “coke”)
  - hybrid-dimension assoc. rules (*repeated predicates*)

    age(X,”19-25”) $\wedge$ buys(X, “popcorn”) $\Rightarrow$ buys(X, “coke”)

- Categorical Attributes: finite number of possible values, no ordering among values—data cube approach
- Quantitative Attributes: Numeric, implicit ordering among values—discretization and clustering approaches

Mining Frequent Patterns II

- FP-Growth: A Frequent Pattern-Growth Approach
- Multi-level Association Rules Mining
- Multi-dimensional Association Rules Mining

Have a question?

